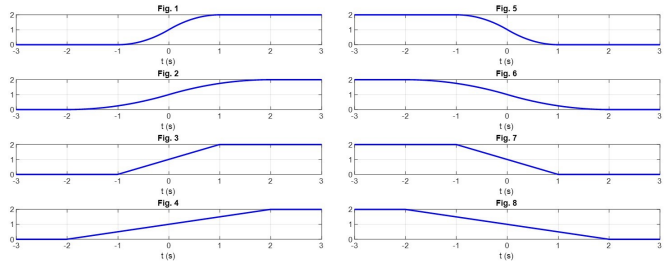
Signals and Systems

Grade Test: Quiz 2

Test Information

QUESTION 1: MULTIPLE CHOICE — **10** **out of 10 points**

Which one of the below figures corresponds to the signal 0.5rect(t/4)*u(-t)?



Given Answer: Fig. 8

Correct Answer: Fig. 8

QUESTION 2: CALCULATED FORMULA — **5** **out of 5 points**

If rect(t/9)*rect(t/9)=k tri(t/9), what is the value of the constant k?

Given Answer: 9

Correct Answer: 9

QUESTION 3: CALCULATED FORMULA | **10** **out of 10 points**

Let f(t)=rect((t-12)/2) and g(t)=tri(t/5).

The signal f(t)*g(t) is nonzero over the time interval a<t<b. What is the value of a?

Given Answer: 6

Correct Answer: 6.0

QUESTION 4: MULTIPLE CHOICE | **10** **out of 10 points**

What is the value of the following integral?

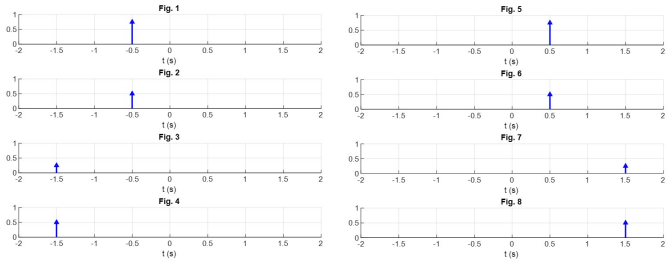
$$\int_{-3}^{+6} \mathrm{tri}\left(\frac{t}{4}\right)\mathrm{comb}\left(\frac{t}{2}\right)\mathrm{d}t$$

Given Answer: 4

Correct Answer: 4

QUESTION 5: MULTIPLE CHOICE | **10** **out of 10 points**

Which one of the below figures corresponds to the signal $2\mathrm{tri}\left(\frac{t}{2}\right)\delta(2t+1)$?



Given Answer: Fig. 1

Correct Answer: Fig. 1

QUESTION 6: CALCULATED FORMULA 10 **out of 10 points**

Let x(t)=4tri(t/2) and y(t)=2rect(t/10).
What is the maximum value of x(t)*y(t)?

Given Answer: 16

Correct Answer: 16

QUESTION 7: CALCULATED FORMULA 10 **out of 10 points**

If δ(t+2)*δ(t+10)*δ(t-4)= δ(t+k), what is the value of the constant k?

Given Answer: 8

Correct Answer: 8

QUESTION 8: CALCULATED FORMULA 10 **out of 10 points**

The function $a\cos(\pi t)\delta(-a(t-b))$ can be written as $c\delta(t-b)$.
What is the value of c if a=9 and b=3?

Given Answer: -1

Correct Answer: -1

QUESTION 9: CALCULATED FORMULA 10 **out of 10 points**

What is the duration of the signal rect((t-4)/4)*tri((t-5)/5)?

Given Answer: 14

Correct Answer: 14

QUESTION 10: CALCULATED FORMULA 10 **out of 10 points**

- Consider the signal x(t) that is a polynomial of order n for all values of t.
- Let y(t)=tri(t)u(t).
- Let z(t)=x(t)*y(t).

If z(t) is a polynomial of order 5, what is the value of n?

Given Answer: 3

Correct Answer: 3

QUESTION 11: CALCULATED FORMULA

5 out of 5 points

What is the period of comb(t/6)?

Given Answer: 6

Correct Answer: 6

FEEDBACK AND NOTES FOR ATTEMPT

Feedback to Learner

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0

Grading Notes

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0